Software Engineering
Assignment-4
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following are desirable properties of a good Software Requirements Specification (SRS) document?

a. It should be concise but not ambiguous.
b. It should specify how the system will be implemented.
c. It should clearly state what the system must do.
d. It should be easy to modify and well-structured.
e. It should be consistent and complete.
f. It should contain contradictory requirements for flexibility.
g. Which of the following are the focus of the requirements analysis activity?

Correct Answer: a. It should be concise but not ambiguous
c. It should clearly state what the system must do.
d. It should be easy to modify and well-structured.
e. It should be consistent and complete

Detailed Solution:

A good SRS must be concise yet unambiguous, clearly state what the system must do (not how), be easy to modify, well-structured, consistent, and complete. It should not specify implementation details (b) or contain contradictory requirements (f)—those are undesirable.

QUESTION 2:

Which of the following are examples of nonfunctional requirements in an MIS software system?

a. The maintenance costs per year should be low
b. The system should be easily portable to other computing environments
c. System should be able to complete the generation of the payroll report within a minute
d. The system should be secure
e. System should take care of the approved leave details of an employee while generating the payroll report
f. Generating monthly payroll reports

Correct Answer: a. The maintenance costs per year should be low
b. The system should be easily portable to other computing environments
c. System should be able to complete the generation of the payroll report within a minute
d. The system should be secure

Detailed Solution:

Nonfunctional requirements specify quality attributes or constraints—low maintenance cost (a), portability (b), performance (c), and security (d) are all nonfunctional. Options (e) and (f) describe specific system functions (integrating leave details, generating reports), which are functional requirements.

QUESTION 3:

Which of the following are sections/subsections are not commonly included in an SRS document as per the IEEE 830-1998 Software Requirements Specification (SRS) standard?

a. Purpose
b. Scope
c. Coding Standards
d. Specific functional requirements
e. Specific nonfunctional requirements
f. Source Code Listings

Correct Answer: c. Coding Standards
f. Source Code Listings

Detailed Solution:

The IEEE 830-1998 SRS standard includes sections like Purpose, Scope, Specific Functional Requirements, and Specific Nonfunctional Requirements—but not Coding Standards or Source Code Listings, which belong to implementation and project documentation, not to requirements specification.

QUESTION 4:

Which of the following characteristics may indicate a poorly written SRS document?

a. Use of customer's terminology to describe functionalities
b. Inclusion of unquantifiable terms such as "good user interface"
c. Use of forward references where concepts are defined later in the text
d. Clearly measurable and testable requirements
e. Wishful thinking about features for which realistic solutions are difficult
f. Consistent and well-structured requirements

Correct Answer: b. Inclusion of unquantifiable terms such as "good user interface"
c. Use of forward references where concepts are defined later in the text
e. Wishful thinking about features for which realistic solutions are difficult

Detailed Solution:

A poorly written SRS contains subjective/unquantifiable terms like "good user interface" (b), forward references to concepts defined later (c), and wishful thinking about unrealistic features (e). Using customer terminology (a), having measurable/testable requirements (d), and being consistent/well-structured (f) are signs of a good SRS.

QUESTION 5:

Consider the following requirement for an online banking system: "The system should allow customers to transfer funds between accounts." Which of the following correctly classify this requirement?

a. Functional requirement
b. Non-functional requirement
c. Requirement describing a service provided by the system
d. Constraint on implementation
e. User interface requirement

Correct Answer: a. Functional requirement
c. Requirement describing a service provided by the system

Detailed Solution:

The requirement describes a specific function (transferring funds) and a service the system provides to users, making it a functional requirement. It is not a non-functional requirement (quality attribute), implementation constraint, or UI-specific requirement, as it does not specify how the interface should look or how the system must be built.

QUESTION 6:

Consider the following requirement for an e-commerce software system: "The application must run on the Linux operating system." It can be considered to be which of the following types of requirements?

a. Functional requirement
b. Non-functional requirement
c. Design constraint
d. Reliability requirement
e. External interface requirement
f. Requirement specifying a business rule

Correct Answer: b. Non-functional requirement
c. Design constraint

Detailed Solution:

The requirement specifies a constraint on the operating environment ("must run on Linux"), which is a non-functional requirement and specifically a design constraint—it restricts design/implementation choices. It is not a functional requirement (it describes no specific behavior), reliability requirement (no uptime/MTBF specified), external interface (no API/hardware interaction), or business rule (no policy/logic).

QUESTION 7:

Which of the following statements regarding SRS review are correct?
a. SRS review is carried out by developers along with user representatives.
b. The purpose of SRS review is to verify that the SRS conforms to actual user requirements.
c. SRS review helps detect defects early and correct them.
d. SRS review is usually performed using checklists.
e. SRS review is mainly intended to generate source code automatically.
f. SRS review is performed only after software implementation is completed.

Correct Answer: a. SRS review is carried out by developers along with user representatives.
b. The purpose of SRS review is to verify that the SRS conforms to actual user requirements.
c. SRS review helps detect defects early and correct them.
d. SRS review is usually performed using checklists.

Detailed Solution:

SRS review involves developers and user representatives to verify that the document matches actual user needs, detects defects early, and often uses checklists for systematic checking. It is not for generating code (e), and it is performed before implementation (f)—early defect detection is the whole point.

QUESTION 8:

Which of the following issues are typically addressed during the requirements analysis activity in software development?

a. Inconsistency in requirements
b. Use of layered design during implementation
c. Incompleteness of requirements
d. Lack of modular coding standards
e. Unstructuredness in requirements
f. Redundancy and unnecessary details in requirements

Correct Answer: a. Inconsistency in requirements
c. Incompleteness of requirements
e. Unstructuredness in requirements
f. Redundancy and unnecessary details in requirements

Detailed Solution:

Requirements analysis focuses on refining and validating requirements—resolving inconsistencies, filling gaps, removing redundancies, and structuring them clearly. It does not address design decisions like layered design (b) or coding standards (d)—those belong to design and implementation phases.

QUESTION 9:

Consider the following partial requirement for a hospital management software system: "The system shall authenticate users using biometric credentials during login." It can be considered to be which of the following types of requirements?

a. Functional requirement
b. Non-functional requirement
c. Security requirement
d. External interface requirement
e. Performance requirement

Correct Answer: b. Non-functional requirement
c. Security requirement

Detailed Solution:

Biometric authentication is a security requirement (a quality attribute under non-functional), and thus also a non-functional requirement (since security is a quality attribute). It is not an external interface requirement (no API/hardware protocol specified) nor a performance requirement (no speed/throughput metric given).

QUESTION 10:

Which of the following questions are commonly included in an SRS review checklist?

a. Have response times been specified for system functions?
b. Have all hardware, external software, and data interfaces been defined?
c. Is each requirement testable?
d. Have the dependencies among modules clearly defined?
e. Are responses to exceptional conditions specified?
f. Has the programming language for implementation been finalized?

Correct Answer: a. Have response times been specified for system functions?
b. Have all hardware, external software, and data interfaces been defined?
c. Is each requirement testable?
e. Are responses to exceptional conditions specified?

Detailed Solution:

An SRS review checklist verifies that performance (response times), interfaces, testability, and exceptional-condition responses are properly specified—all essential for a complete, unambiguous requirements document. It does not include module dependencies (d) or programming language (f), as those belong to design and implementation decisions, not to the SRS itself.

******THE END******